

Original Article

Small-scale sea urchin (*Tripneustes gratilla*) aquaculture: evaluating locally sourced diets for gonad development and quality

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Abstract

Sea urchin gonads (called uni) are highly valued in seafood markets, creating demand for sustainable production. In Seychelles, small-scale farming of *Tripneustes gratilla* is promising, but costly pelleted feeds for gonad enhancement limit feasibility. This study evaluated alternative fresh feeds, *Sargassum spp.*, carrots (*Daucus carota*), moringa (*Moringa oleifera*), and a mixed diet, over 10 weeks using 200 *T. gratilla* (average wet weight: 492.6 ± 79.37 g). Gonad quality was assessed at weeks 6, 8, and 10, alongside disease and mortality monitoring. The overall mean gonadosomatic index (GSI) was 3.34 %, peaking at 10.38 % in week 8. Significant differences were found among feed types and durations ($p < 0.05$). Moringa had a strong negative effect on GSI (-3.54, $p = 0.001$), while other diets showed no significant difference. GSI rose from week 6 to 8 before declining at week 10 (estimate = -2.18, $p = 0.035$). However, 46.67 % of gonads were of unacceptable quality. Feed type and duration had no significant impact on gonad colour ($p > 0.113$), texture ($p > 0.071$), or firmness ($p > 0.064$). Mortality and disease rates were also unaffected ($p > 0.415$ and $p > 0.963$). Although no tested diet improved gonad quality, the potential for artisanal *T. gratilla* farming may remain, but likely depends on improved enhancement diets and/or locally feasible formulated feeds, which require further evaluation.

Keywords: *Tripneustes gratilla*, echinoculture, SIDS, artisanal aquaculture, aquaculture nutrition, sustainability

Introduction

Sea urchin gonads, commonly known as uni, are a prized delicacy in global cuisine, especially in high-end Japanese and fusion dishes (Sun and Chiang, 2015). Their creamy texture and distinctive, delicate flavour have spurred rising demand from gourmet markets, making uni a high-value product in the international seafood trade (Raposo *et al.*, 2019; Sun and Chiang, 2015). This demand, however, places immense pressure on wild stocks, many of which are now overexploited (Sun and Chiang, 2015). Consequently, there is growing interest in sustainable aquaculture solutions to ensure a reliable, high-quality

supply of uni that meets both market expectations and environmental goals.

One of the key challenges currently facing echinoculture is the lack of a sustainable and effective diet that can enhance gonad production while maintaining desirable nutritional and sensory qualities (Ciriminna *et al.*, 2020). More recently, Raposo *et al.* (2019) also advocated for the use of terrestrial vegetables as alternative feed sources, noting their potential to support both gonad growth and favourable fatty acid profiles in sea urchins, in place of cultivated macroalgae or commercial feeds.

Access to affordable and effective feed for gonad enhancement presents a crucial barrier to scalable, low-cost production. High-quality gonads are critical for uni markets, and formulated feeds with 20 % *Ulva lacunulata* inclusion have shown strong results in increasing gonad mass (Cyrus *et al.*, 2015, 2014), although their practicality in SIDS contexts is limited. These feeds are costly comprising 50 to 70 % of total operational expenses (Vassiliou *et al.*, 2015) and require cold storage, which can be a logistical challenge in tropical environments. Additionally, they are not locally produced and are prone to spoilage and contamination (de Vos *et al.*, 2024; Shpigel *et al.*, 2018). This makes the search for locally sourced, low-cost alternative feeds for sea urchin aquaculture a pressing priority.

Aquaculture in Seychelles is a relatively recent development, with the Seychelles Fisheries Authority serving as the primary regulatory and oversight body. The Authority has identified several candidate species with high aquaculture potential and is currently conducting research to establish optimal culture methodologies for these species. To date, prawns remain the only species cultured at a commercial scale in Seychelles.

Tripneustes gratilla is considered a promising aquaculture candidate due to its fast growth, palatability, and favourable farming characteristics. This species is indigenous to the Seychelles and has been chosen as one of its key species for aquaculture production (Seychelles Aquaculture, n.d.). The islands' stable tropical climate, free from cyclonic disruptions, and consistent sea temperatures averaging 27 to 32 °C (Heubeck 2022) create ideal conditions for sea urchin growth and reproduction (Seychelles Aquaculture, n.d.). In addition, Seychelles offers vast areas of shallow, sheltered marine environments suitable for low-tech farming systems such as cages and pens (Lee *et al.*, 2023; Mackie *et al.*, 2005). These conditions allow for year-round production, which is critical for maintaining consistent supply chains.

Locally available feed alternatives offer a potential solution. This includes *Sargassum* spp., a brown seaweed that grows abundantly and is readily consumed by *T. gratilla* (personal observation, 2022). Studies show *Sargassum*-fed urchins exhibit good growth rates (SGR up to 1.62 mm), particularly in juveniles, but achieve only moderate gonad development (GSI ~5 %) (Mos *et al.*, 2012; Juinio-Menez *et al.*, 2008). This suggests that even though *Sargassum* may support somatic growth, it may not be optimal for gonad enhancement alone.

Carrots (*Daucus carota*) offer another intriguing possibility. Rich in carotenoids, they may improve gonad colour, a key quality factor in uni (Pinheiro-Sant'ana *et al.*, 1998; Plank *et al.*, 2002). Although they have not been studied in *T. gratilla* trials, in *Strongylocentrotus droebachiensis* they showed neutral effects on gonad quality (Robinson *et al.*, 2002). Their local availability and affordability make carrots a viable subject for further investigation in tropical aquaculture systems.

Another potential feed is *Moringa oleifera*, widely available in Mahé. It is known for its nutrient-dense leaves and high carotenoid content (Haroen *et al.*, 2022), and is already consumed across many cultures. While there are no published studies on its use in sea urchin diets, preliminary trials suggest that *T. gratilla* readily consumes moringa leaves (personal observation, 2023). Given its accessibility and nutritional value, moringa could serve as a sustainable, low-cost ingredient in sea urchin feed formulations.

Although *Ulva* spp. is a proven enhancer of gonad quality, it presents cultivation and supply challenges in small-island contexts. Large-scale *Ulva* farming requires significant land and infrastructure investments, which may not be feasible for artisanal farmers. Wild *Ulva* harvesting in Seychelles is possible but limited. Therefore, instead of relying solely on *Ulva*, a more realistic solution may lie in developing a mixed diet incorporating fresh *Sargassum*, moringa, carrots, and small amounts of *Ulva* where available. Such mixed diets have shown promise in other studies (Brink *et al.*, 2022), yielding improved gonad quality over single-source feeds.

The objective of this research is to contribute to food security and economic diversification in island communities by advancing sustainable and decentralized approaches to sea urchin aquaculture, thereby aligning with global efforts toward resilient and environmentally responsible seafood production. Exploring alternative, locally sourced feeds have important implications for sea urchin aquaculture in Seychelles and other SIDS. This study addresses the need for affordable, scalable feeding strategies that avoid the costs and limitations of imported feeds. By assessing the effects of carrots, moringa, *Sargassum*, and mixed diets on the gonad colour, texture, firmness, and gonadosomatic index of *T. gratilla*, this study aims to support small-scale aquaculture development.

Materials and methods

Experimental design

This experiment was conducted at the Sea Urchin Research Facility (SURF), located in the Seychelles Maritime Academy in Providence, Mahé, Seychelles in a land-based flow-through system to standardise environmental conditions and isolate dietary effects. *Tripneustes gratilla* were held in crates (50 cm L × 32 cm W × 26.5 cm H) that were suspended from wooden beams to keep them half-submerged, covered with plastic sheets for additional shading, and placed in fiberglass raceways (6 × 1.8 × 0.8 m; 8.5 m³ volume). These raceways were supplied with constant flow-through seawater, which filtered through a sand filter. The flow rate was approximately 0.5 turnovers per hour. The tanks were constantly aerated and, in addition to being covered with a 50 % shade cloth, were exposed to ambient environmental conditions throughout the two-month duration of the experiment. A 14-day acclimation period was conducted prior to the experiment, during which the sea urchins were exclusively fed *Sargassum* spp. every other day; during this period, 14 cases of bald spot disease were observed, but no mortalities occurred. Throughout the study, dissolved oxygen levels of 5.45 mg/l to 5.97 mg/l and water temperature (°C) 26.8 °C to 30.2 °C were monitored daily using a portable HQ series DO meter model HQ1130 (with precision rating of 0.1 mg/l and 0.1 °C). Additionally, to ensure water quality remained optimal, the tank bottoms were siphoned of debris and faecal matter every other day. Daily inspections were conducted to identify and record any instances of disease or mortalities. In the event of such occurrences, affected sea urchins were removed from the system and recorded.

Feeding trial

A total of 20 baskets were used during the experiment, with each basket containing 10 *Tripneustes gratilla*. Four dietary treatments were tested: moringa, carrot, a mixed diet, and *Sargassum* spp. Therefore, there were five replicates (baskets) per feed treatment. Urchins fed moringa, *Ulva* and *Sargassum* spp. were provided feed *ad libitum*, as this would not decay if not consumed by

the urchins. Carrots, on the other hand, were supplied every second day at 10 % of the total body weight of the urchins in each crate, with adjustments made for mortalities or disease to maintain accurate feeding rates. This carrot feeding schedule was implemented because the uneaten carrots would begin to decay after two days but generally the urchins would consume them at a rate of 10 % of their body mass over this two-day period. Any carrot not eaten would be removed after two days. This was seen from preliminary trials that was conducted prior to the experiment. The mixed diet alternated weekly among moringa, carrot, *Ulva* spp., and *Sargassum* spp., The feeding frequency and quantity was the same as for each unmixed diet, meaning, moringa, *Ulva* spp., and *Sargassum* spp. were fed *ad libitum* while the carrots were fed at 10 % body mass of the urchins (Table 1). This was done so that *T. gratilla* always had food available to them. Additionally, it is essential to recognize that feeding rates observed in natural environments may differ significantly from those in aquaculture settings, depending on the composition and nutritional quality of the diets provided (Bayed *et al.*, 2005). In experiments evaluating different feeding rates and periods, results indicated that continuous feeding typically leads to superior growth and gonad development compared to intermittent feeding regimes (Cuesta-Gomez and Sánchez-Saavedra, 2018; Zuo *et al.*, 2022)

Gonad analysis

Gonad qualities were assessed through dissections conducted on three occasions: firstly, on the 16th of October during the waxing crescent phase; secondly, on the 27th of October during the waxing gibbous phase; and lastly on the 10th of November during the waning crescent phase. This was necessary to determine if moon phases affected the quality of *T. gratilla* gonads.

A total of 40 *T. gratilla* were dissected during each dissection event, with two individuals randomly handpicked from each basket (n = 10 per treatment). Gonad colour, texture, and firmness were evaluated using the visual classification system adapted from Pearce *et al.* (2002).

Table 1. Treatments fed to *Tripneustes gratilla* during the experiment and their corresponding frequency and rate.

Diet	Feeding frequency	Feeding rate
Moringa	Daily	<i>Ad libitum</i>
Carrot	Every 2 days	10% of total body weight
Mixed diet	Depending on feed type, alternated each week	Depending on feed type
<i>Sargassum</i> sp.	Daily	<i>Ad libitum</i>

Gonad colour

Eye-rated gonad colour categories included: (1) bright yellow-orange gonads (excellent quality); (2) yellow-orange gonads (acceptable quality); (3) pale yellow-orange or dark yellow-orange gonads (low quality); and (4) white or brown gonads (unacceptable).

Gonad texture

For texture, the following criteria were used: (1) two distinct gonad segment halves, very smooth; (2) two distinct gonad segment halves, smooth (distinction and smoothness <1); (3) distinction of gonad segment halves possible but < 2, rough/granular; (4) distinction of gonad segment halves not possible, rough/granular.

Gonad firmness

Gonad firmness was based on the following criteria: (1) very firm; (2) firm; (3) soft; (4) very soft.

Values within each dietary treatment were averaged to generate a single value for comparison between treatments. A lower average eye-rated value for colour, firmness, and texture indicated superior coloration and thus higher gonad quality.

Gonadosomatic index

The body (W_b) and gonad (W_g) wet weights of individual urchins with precision of 0.3 g and 0.1 g respectively (de Vos *et al.*, 2023) fed the various experimental diets

were used to calculate gonadosomatic index according to the following formula used by Pearce *et al.* (2002):

$$\text{GSI (\%)} = W_g / W_t \times 100$$

Statistical analysis

All results are presented as mean $\pm$ standard deviation (SD) to describe central tendency and variability among replicates within each treatment. The assumptions underlying the statistical models were thoroughly assessed. The absence of missing data points eliminated concerns related to missing values, ensuring the integrity of the dataset. To verify the suitability of linear mixed-effects models (LMMs), a suite of diagnostic plots was employed. Linearity and normality of residuals were evaluated using quantile-quantile (QQ) plots (Supplementary Fig. A), while the residuals versus fitted values plot (Supplementary Fig. C) was used to assess homoscedasticity. Model fit was examined by comparing observed versus predicted values (Supplementary Fig. B). These diagnostics (Supplementary Figs. 6 to 9) confirmed that the assumptions of normality, linearity, and homogeneity of variance were met, and that the models were appropriate for detecting significant treatment effects.

A linear mixed model was used to assess the impact of feed type and duration on the gonadosomatic index (GSI) in *T. gratilla*. The main statistical approach

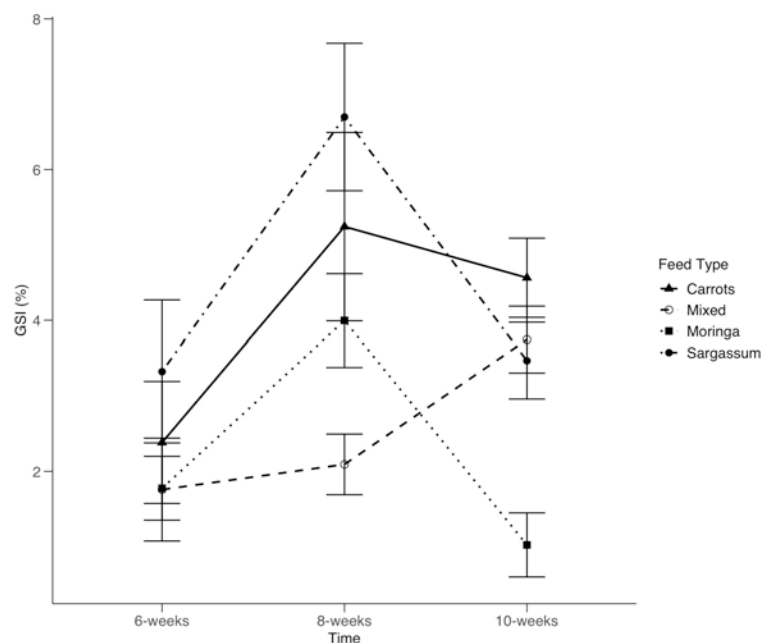


Figure 1. Variation in the Gonadosomatic Index (GSI) of *Tripneustes gratilla* over three-time intervals (6 weeks, 8 weeks, and 10 weeks) across different feed types. Error bars represent standard deviation. The average GSI (%) is shown for each feed type: *Sargassum* (grey), Carrots (orange), Moringa (dark green), and Mixed diet (brown). A significant increase in GSI was observed from 6 to 8 weeks, followed by a decline at 10 weeks. Moringa exhibited a negative effect on GSI compared to other diets.

involved the use of LMMs to analyse variation in gonad size index (GSI), colour, texture, and firmness. These models were chosen to account for the repeated measures design, in which the same baskets were assessed over time, and to address potential issues of non-independence and pseudoreplication. Fixed effects included feed type, sampling time, and their interaction, while random effects accounted for variability among baskets (Schielzeth *et al.*, 2020).

Disease incidence and mortality rates were analysed separately using Kruskal-Wallis tests, as the Shapiro-Wilk test indicated that the data for both variables were not normally distributed ($p < 0.001$). However, Levene's test confirmed that the assumption of homogeneity of variances was satisfied ($p > 0.697$), justifying the use of non-parametric tests.

All statistical analyses were performed using R programming language, version 2024.042 + 764.

Results

Gonadosomatic Index

The overall average ($\pm$ standard deviation) GSI across all treatments was (3.34 ± 2.63 %), with a maximum observed value of 10.38 %. Four individuals exhibited no gonads (3.3 %). The model revealed significant effects of specific feed types and time intervals on GSI ($p < 0.05$).

Moringa exhibited a significant negative effect on GSI compared to other feeds (estimate = -3.54 , $p = 0.001$), whereas the remaining feed types did not significantly differ from each other ($p > 0.286$). GSI increased from week six to week eight but declined at week 10 (Fig. 1). This trend was statistically supported by a significant negative coefficient for six weeks (estimate = -2.18 , $p = 0.035$).

Eye-rated gonad colour

Two out of 120 individuals (1.6 %) exhibited excellent-quality gonads (bright yellow-orange, score 1). A total of 22.5 % of urchins had acceptable quality gonads (yellow-orange, score 2), while another 22.5 % were classified as low quality (pale yellow-orange or dark yellow-orange, score 3). The majority (46.67 %) displayed gonads that were white or brown (score 4), rendering them unsuitable for market standards.

There was a slight trend suggesting that *Sargassum*-fed urchins had marginally better colour at weeks 8 and 10 (Fig. 2). However, the linear mixed model analysis indicated no significant influence of feed type or enhancement duration on gonad colour ($p > 0.113$).

Gonad texture

Five out of 120 individuals (4.2 %) exhibited high-quality gonad texture (score 1), distributed across all feed treatments except for *Sargassum*. All instances of

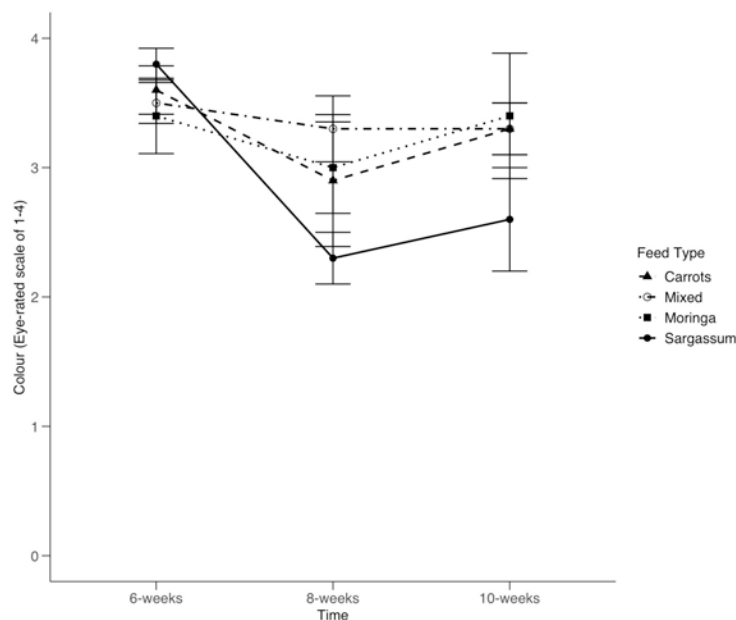


Figure 2. Effect of different feed types on gonad colour of *Tripneustes gratilla*. Higher scores on the colour scale correspond to poorer gonad quality. Bright yellow-orange gonads (score 1) -excellent quality, yellow-orange gonads (score 2) -acceptable quality. Pale yellow-orange or dark yellow-orange gonads (score 3) -lower quality, white or brown gonads (score 4) -unacceptable. Error bars represent the standard error.

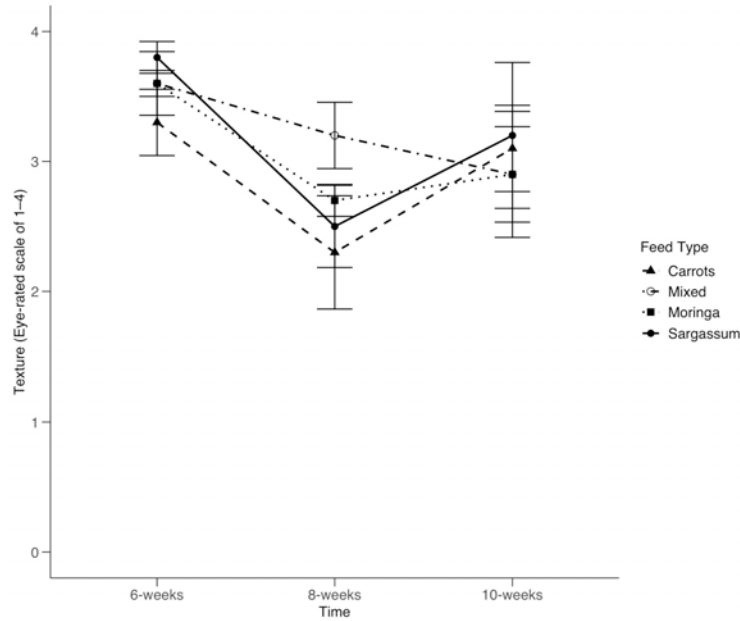


Figure 3. Eye-rated gonad texture, depicting the mean values for each feed type with error bars representing standard errors. The texture scores range from 1 to 4, where (1) two distinct gonad segment halves had little to no follicle separation, (2) two distinct gonad segment halves with visible follicle separation, (3) distinction of gonad segment halves is possible but quite granular, and (4) distinction of gonad segment halves is not possible, with gonads appearing rough or granular.

high-quality texture were observed after eight weeks of feeding. The overall average texture improved at week eight but declined by week 10 (Fig. 3). However, feed type, time, or their interactions did not significantly affect gonad texture ($p > 0.071$, Table 4). Overall, 46.67 % of gonads received the lower texture score of 4, indicating poor quality.

Gonad firmness

Gonad firmness did not show clear patterns of variation based on diet or time (Fig. 4). Only one individual (0.8 %), fed with *Moringa*, exhibited a very firm gonad (score 1) and had a GSI of 7.75 %. Among the other urchins, 39 (32.5 %) were classified as firm (score 2), 34 (8.3 %) as soft (score 3), and 46 (38.3 %) as very

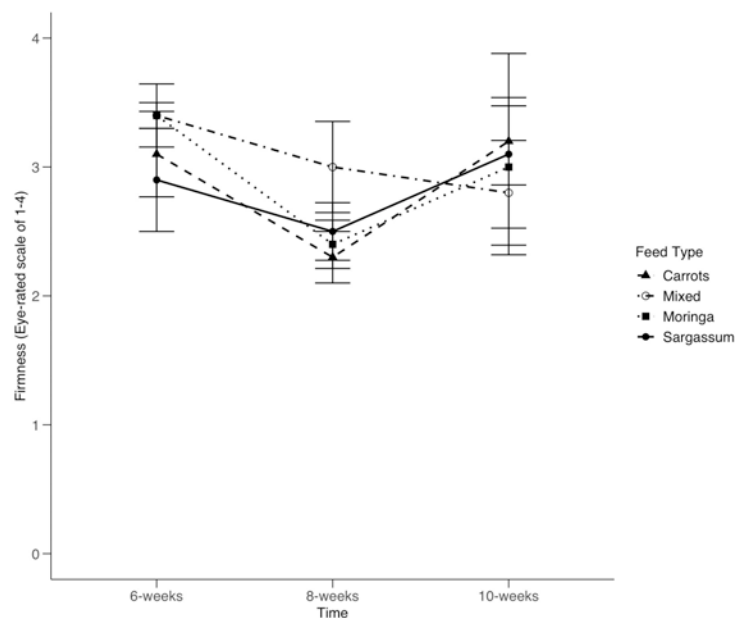
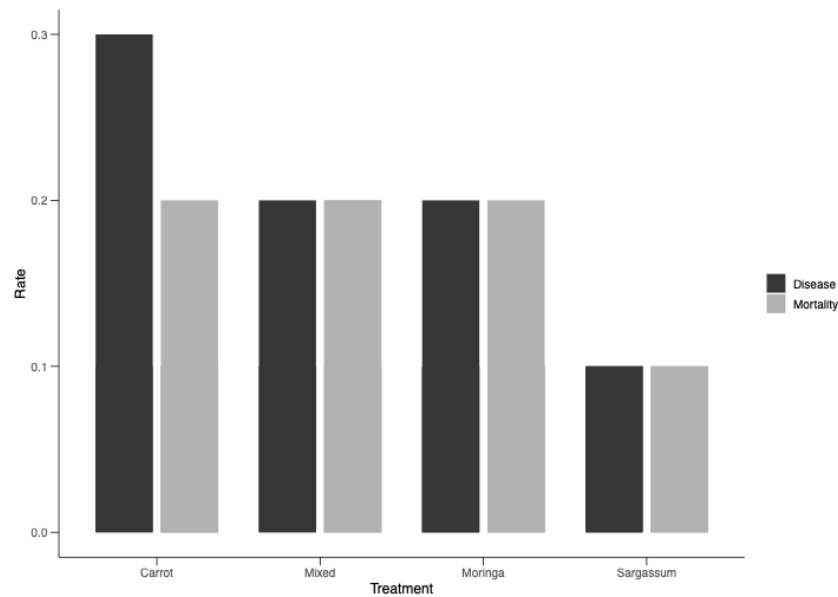


Figure 4. Eye-rated firmness of sea urchin gonads over time based on various feed types, categorized on a scale from 1 to 4: (1) very firm; (2) firm; (3) soft; and (4) very soft. Each line represents a different feed type, with error bars indicating standard errors.



Figures 5A and 5B. The graph illustrates the variation in disease (A) and mortality rates (B) among various treatment conditions, providing insights into the effectiveness of different approaches for enhancing sea urchin gonads. Treatments are represented on the x-axis, while the y-axis indicates the respective rates, with error bars indicating standard errors.

soft (score 4), making them unsuitable for marketability. Linear mixed model analysis found no significant effects of diet, time, or their interactions on gonad firmness, except for a marginal effect observed at week 8 (estimate = -0.90, $p = 0.064$).

Disease and mortality rates

An average ($\pm$ standard deviation) mortality rate of 0.8 ± 0.77 individuals per basket was recorded across all treatments (10 individuals per basket), with a range of 0 to 3 mortalities. The *Sargassum*-fed group exhibited slightly lower average mortality rates (Fig. 5A), but no significant differences were detected between dietary treatments (Kruskal-Wallis test, $\chi^2 = 2.85$, $df = 3$, $p = 0.415$). Disease incidence ranged from 0 to 3 cases per basket (Fig. 5B), with no significant variation among treatments ($\chi^2 = 0.284$, $df = 3$, $p = 0.963$). Thus, dietary treatments had no significant effect on disease or mortality rates in *T. gratilla*.

Discussion

The aim of this study was to evaluate the efficacy of locally available feeds (carrots, moringa, *Sargassum*, and mixed diets) on the gonad quality (colour, texture, firmness, and gonadosomatic index) of *Tripneustes gratilla*, thereby supporting small-scale sea urchin aquaculture in the Seychelles and other Small Island Developing States (SIDS). The results of this study reveal that none of the tested diets significantly improved gonad quality.

Despite moringa's reputation as a nutrient-rich feed source due to its high carotenoid content (Haroen *et al.*, 2022), *T. gratilla* fed with moringa exhibited the lower GSI. This outcome may be attributed to the physical properties of moringa leaves, which were fragmented and lost through basket slits, potentially reducing effective intake. *T. gratilla* have been observed to consume moringa, and the inability to quantify actual consumption presents a limitation in assessing its impact on gonad development. No statistically significant differences in GSI were observed among urchins fed mixed diets, *Sargassum*, or carrots. The gonadosomatic index showed a notable temporal pattern, increasing from week six to week eight and then declining at week 10. This trend aligns with the lunar cycle, where GSI peaked approaching the full moon, suggesting spawning occurred thereafter, leading to reduced gonad mass. Similar lunar phase influences on reproductive cycles have been reported in *T. gratilla* (Johnson and Ranelletti, 2017; Muthiga, 2005), highlighting the importance of considering lunar periodicity when designing feeding experiments. Such studies have shown a significant relationship between gonad index and lunar phases in *Tripneustes gratilla*, with spawning occurring between lunar days 7 and 21 (Muthiga, 2005). However, since a full gonad maturity cycle was not observed during the study period, it remains uncertain whether its reproductive cycle aligns with the lunar cycle, highlighting the need for long-term research. A more recent study

by Hernandez *et al.* (2022) on the sea urchin *Arbacia punctulata* found that males developed sperm in sync with the new moon in June and July. Some individuals retained residual sperm (~7 %), while nearly half (~46 %) had testes fully occupied by mature sperm. Females also displayed a pattern, with many having mature eggs (~39-71 %) or empty follicles around the new moon, suggesting that spawning occurs during this period. Cunningham *et al.* (2023) observations of the mature reproductive stage in *L. variegatus* were linked to the new and full moons, suggesting that the lunar cycle or tides influence spawning timing. Peak spawning periods occurred in spring, with occasional spawning in summer, while spawning likely did not take place from October to January. Additionally, environmental factors such as light intensity may influence gonad development, as *T. gratilla* has demonstrated acclimation to varying light conditions (Payet *et al.*, 2024).

Gonad coloration is a pivotal factor influencing market acceptability, with international markets favouring bright yellow or orange gonads (Cyrus *et al.*, 2013, 2015; McLaughlin and Kelly, 2001; Robinson, 2002). In this study, however, 46.67 % of the dissected urchins displayed white or brown gonads, rendering them commercially unviable. The average test diameter of *T. gratilla* utilized in this study was 92.5 mm, suggesting a potential link between the observed gonad coloration and reproductive maturity. Optimal gonad coloration has been documented during the prematuration and early maturation phases, whereas diminished coloration is typically associated with late maturation, spawning, and recovery periods (Tjendanawangi *et al.*, 2010). Additionally, the dietary bioavailability of carotenoids is crucial for gonad pigmentation, as demonstrated by Tjendanawangi and Dahoklory (2020). The findings indicate that despite moringa and carrots being rich in carotenoids, their bioavailability in *T. gratilla* may have been insufficient to enhance gonad coloration. Previous studies have suggested that ingredients such as carrots and paprika can help maintain gonad pigmentation, however, the concentration of these pigments must be sufficiently high to induce noticeable changes (Robinson *et al.*, 2002). Gonad texture is another essential quality parameter. In this study, gonad texture was deemed market-acceptable at week eight, coinciding with the waxing gibbous moon phase, suggesting a possible lunar influence on gonad texture development. Notably, *Sargassum*-fed urchins exhibited the least favourable texture, a finding inconsistent with previous research by Tjendanawangi and Dahoklory (2020), who reported high gonad texture

quality in urchins fed *Sargassum* within four weeks. This discrepancy may be attributed to differences in experimental duration, seaweed composition, or environmental conditions. Firmness is a critical attribute for gonad marketability, as firmer gonads are preferred by consumers. In this study, only one urchin exhibited gonads with the desired firmness, reinforcing the conclusion that none of the tested diets were optimal for gonad enhancement. However, the diets did not negatively impact urchin health, as no significant differences in mortality or disease incidence were observed. This suggests that while these diets may not be suitable for enhancing gonad quality, they may be viable for urchin growth in aquaculture as they did not adversely affect urchin health. Further grow-out trials are essential to validate this potential application. It may be necessary to conduct an extended experiment to ascertain whether the diet truly does not affect urchin gonads. Additionally, using crates without slits could better evaluate the consumption rate of moringa in relation to *T. gratilla*. While formulated feeds have been shown to produce substantially higher GSI in *T. gratilla*, their cost and cold-chain requirements may limit their suitability for artisanal producers in SIDS. Future work should therefore explore locally manufacturable formulations (e.g., higher-protein diets with effective carotenoid delivery) and assess feasibility through a dedicated techno-economic analysis for both local and export markets. Although environmental impacts were not quantified in this diet trial, shallow-water urchin farming must be evaluated within a carrying-capacity and ecosystem-risk framework.

Conclusion

In conclusion, this study demonstrates that the locally sourced feeds tested; carrots, moringa, *Sargassum*, and mixed diets did not enhance the gonad quality of *T. gratilla* in terms of gonadosomatic index (GSI), colour, texture, and firmness. The findings underscore the potential influence of lunar cycles on reproductive growth, suggesting that feeding experiments should account for these periodic variations. Additionally, the results highlight the importance of carotenoid bioavailability, as the mere inclusion of pigments in feed does not necessarily lead to improved gonad coloration. Although these diets did not adversely affect urchin survival or health, they also failed to promote significant gonad development, indicating that alternative feeding strategies, such as formulated diets or nutrient-rich macroalgae, may be required for effective gonad enhancement. While these diets were not effective for gonad enhancement, feasibility of

small-scale *T. gratilla* farming in Seychelles remains uncertain and will depend on improved enhancement diets and/or practical formulated feeds, together with a dedicated techno-economic assessment.

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